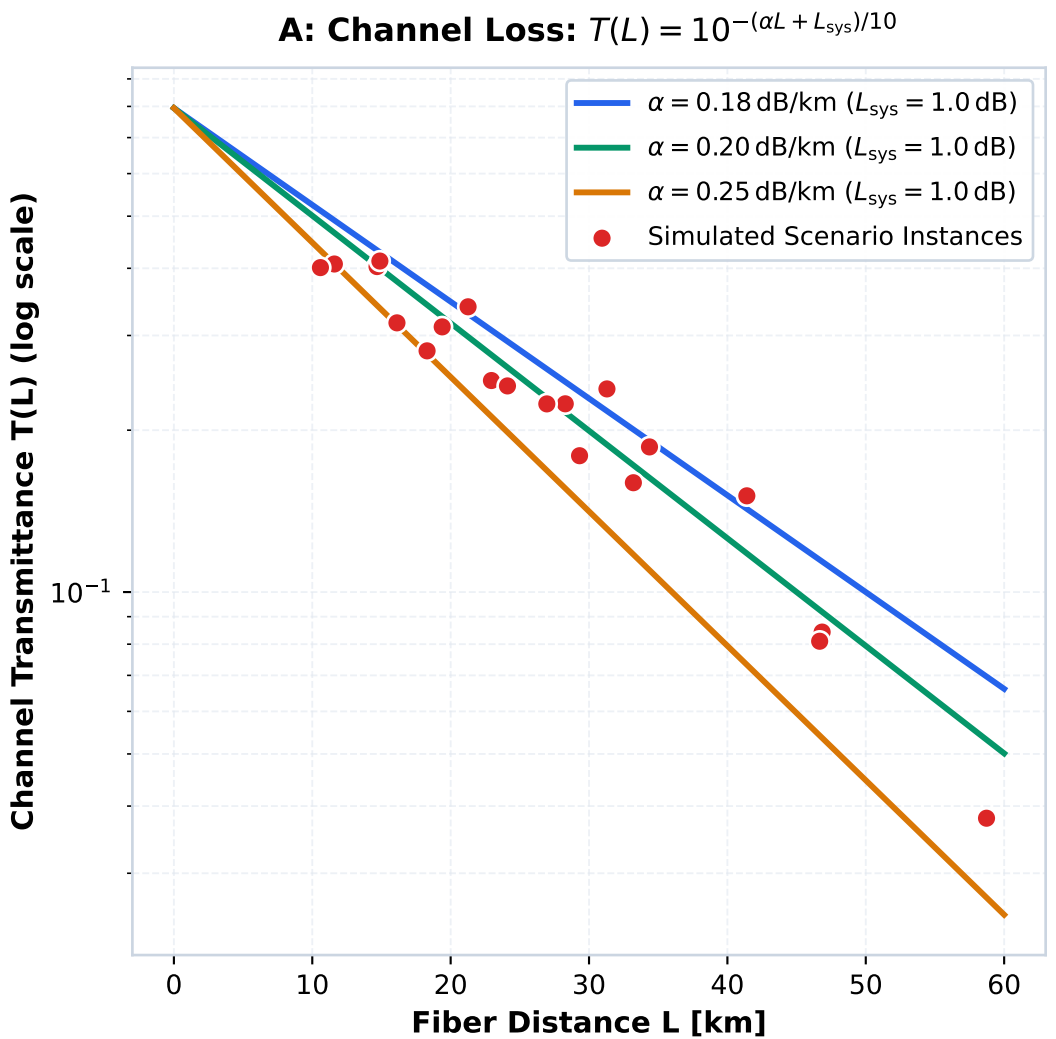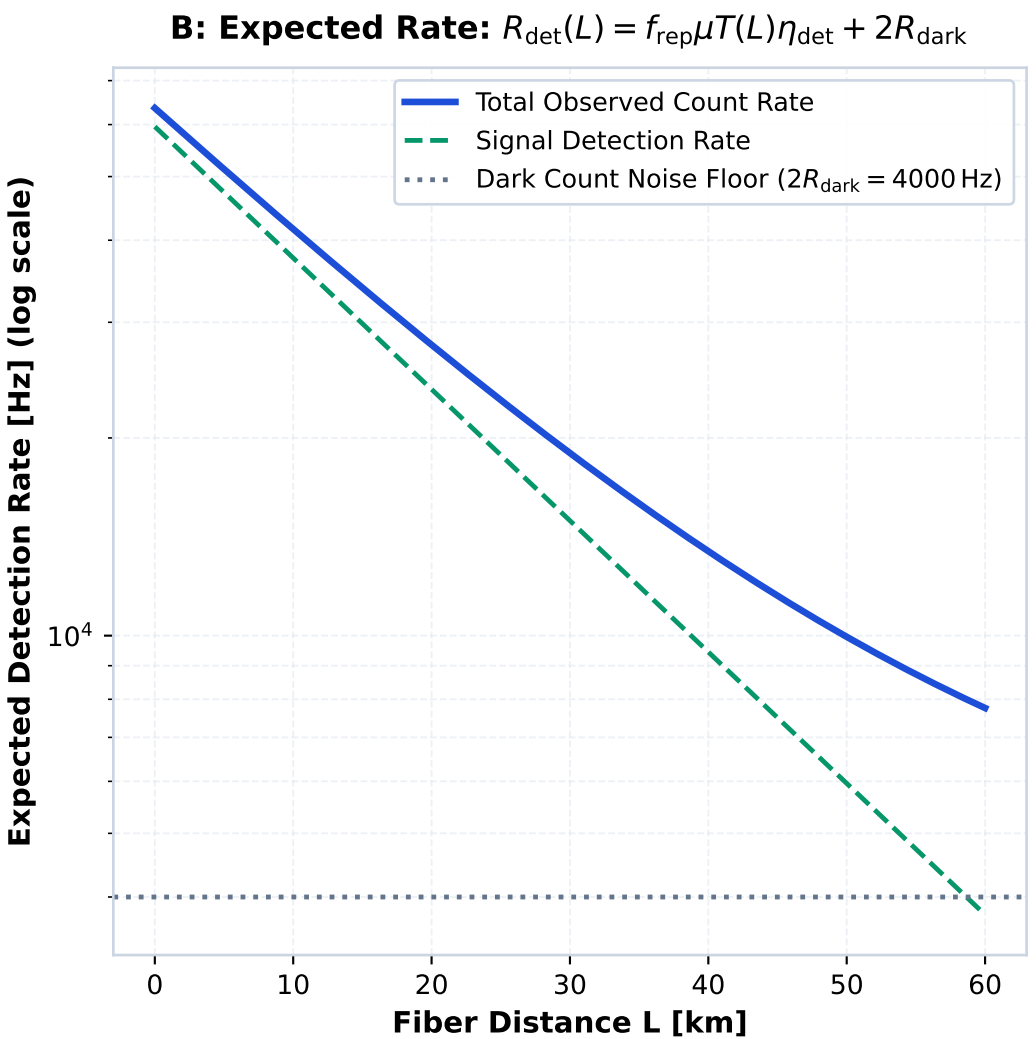Quantum Channel Physics: Optical Attenuation and Dark-Count Noise Floor